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Introduction

1. In this book, I will show that physics, as it has taken shape up to the present, is non-dialectical in character. I will limit my discussion to the period up to approximately 1930. Subsequent physics merely develops the errors of earlier physics.

2. My aim is to show that physics, from the Renaissance through Newton to Einstein, including Einstein himself—all of this physics was built on reducing matter to motion understood as change, rather than reducing matter to development.

3. Einstein's theories did not overcome Newton's errors or misconceptions. They merely manifested the irresolvable contradictions of physics, carrying its fundamental proposition—motion as change—to its conclusion.

4. I also intend to show how dialectical physics can be constructed. By identifying the error of past physics, we can construct dialectical physics as a physics of development, rather than of motion.

5. In studying physics, I became convinced that the way it is taught today, like the way mathematics is taught, is mistaken. Mathematics is just as non-dialectical as physics. I will therefore present an account of dialectical teaching of these and other subjects. Our brain is a dialectician; our teachers are not. This must be corrected. Mathematics and physics themselves may be non-dialectical, but their development is dialectical.

6. By observing how I study physics and mathematics, I have gathered a substantial body of material for creating AI. This must be dialectical AI. The book will therefore discuss an approach to creating such AI. Here, the subject will be the dialectical study of the world, rather than dialectical teaching.

Section 1. Dialectics

1. Dialectics takes as its fundamental proposition that objective reality, that is, matter, is development, motion.

Dialectics is the study of motion, but of motion understood as development.

If development is taken as the starting point, space and time are not required. In this case, the unity of matter means precisely development.

2. Development is the result of the becoming of "the one", which already contains "the other", and the becoming of "the other", which already contains "the one". Becoming is the transition of the one into the other, and of the other into the one.

Development proceeds as follows:

1) everything begins with a simplest process. The simplest process is one that itself arose as the resolution of a contradiction and is itself capable of development;

2) the simplest process develops through other processes. This development proceeds as follows. The simplest process already is what it is, that is, "the one", but also contains what it is not, that is,

“the other”. The simplest process presents itself as the identity of “the one” and “the other”, through which the transition into “the other” takes place. This “other” is called a developing process. Each developing process generates subsequent developing processes, just as the simplest process generated a developing process;

3) the tendency of the generation of developing processes towards infinity constitutes the opposite of the simplest process, since the developing processes are “the other” for the simplest process;

4) taken together in their development, the simplest process and its opposite constitute a contradiction;

5) the development of the contradiction requires its resolution;

6) the resolution of the contradiction, its sublation, takes place through a leap, that is, without the identity of “the one” and “the other”.

3. There is a world of difference between proceeding from ideas to reality and proceeding from reality to ideas.

4. Negation should not destroy; it should be a process that resolves the contradiction and allows it to exist, but on another, higher level. Destroying a seed does not mean negating it, in the sense that a second negation is then impossible: a tree cannot be obtained from the seed.

5. Determinacy is already itself both cause and effect.

6. Dialectical analysis: 1) find the simplest and give its full definition; 2) find what contradicts it; 3) find the resolution of this contradiction.

A contradiction arises where one excludes the other. Does A exclude B? Then we have a contradiction. It is resolved in a C in which either A and B are excluded and replaced by something else, or A or B is not excluded and, on the contrary, A and B can continue to exist, moreover, as contradictions.

To find a contradiction, we must show the development of the simplest towards its limit. Development towards a limit does not mean complete development: it means precisely tending towards a limit without reaching it. The fuller the description of “the one”, the fuller the description of “the other” will be.

Dialectics seeks to derive the complex from the simple, rather than to reduce the complex to the simple. In this sense, Marx took the commodity and showed its motion. We must find the simple, define it, and disclose it fully; this definition will already determine its motion. The search for the simplest is a task, a point of orientation, but everything should not be reduced to it. The simplest must necessarily exist and be found, but what we must seek is motion itself. This principle should also help us discard unnecessary knowledge, retain only that in which motion can be discerned, and transform data so that motion can be discerned in it.

Processes can be considered logically and historically. The result should be the same. Where historical data are insufficient, we rely on logic. Where logic is insufficient, we use historical data. In either case, what must unite these two methods is precisely development, the search for such development.

Non-dialectical logic holds that the universal is only what all objects have in common, whereas dialectical logic holds that the universal is that through whose development everything else became possible. Marx takes the simplest object—the commodity. This simplest object has its own particular characteristics; they are not discarded. The development of the commodity shows how it acquires these characteristics. Non-dialectical logic would take a different route. It would take objects, discard everything specific to them, and retain only what they all have in common.

7. Understanding matter as motion substantially changes our view of many questions.

Will Achilles catch the tortoise? The question matters, but only if you accept space and time, rather than motion. If you recognise motion, there are no Achilles and tortoise as objects. This contradiction does not exist. Let Achilles and the tortoise be a “sphere” of infinity. Then this sphere continually changes its probability. This is precisely the “change” in which Achilles moves towards the tortoise.

8. The difference between simple and dialectical logic. In simple logic, if all humans are mortal and you are human, then you are mortal. In dialectical logic, however, you are both mortal and immortal at the same time. One cannot exist without the other. I am mortal; that is so. I am mortal as this particular shell, this particular person. But I am also immortal as a representative of humanity, of the world, of “nature”. Only through the latter am I human; conversely, everything else is what it is because I am human.

9. According to Engels, the first breach in the metaphysical approach to nature was made by Kant with his Universal Natural History and Theory of the Heavens (1755).

10. Quality is development considered from the side of contradiction. Quantity is development considered from the side of infinity.

Quality is infinity insofar as it is infinity (without boundaries).

Quantity is infinity insofar as it has boundaries.

Subsection 1.1 Mathematics

1. Mathematics can be reconstructed dialectically as follows:

1) a number is a unit. Only the unit is taken to be a number. Moreover, what is recognised as a unit, that is, as a number, is a unity, a “halt” in the process of development. At first, 1 is taken to be indivisible, because it is not understood that the unit is itself a reference. Gradually, it is accepted that the unit is a reference. This is where fractions appear, but the unit can be both a sum of halves and a sum of thirds, which no longer yields an exact solution.

Complex numbers, fractional numbers, and all other kinds of numbers must be considered as operations, specifically as summation or a complication of summation.

One cannot stare at 2*1 and understand that it is 1+1;

2) a number is quality and quantity simultaneously. The very concept of number already contains such concepts as prime number;

3) two, three, and so on are not numbers, but references to operations;

4) an operation is summation and what is derived from it; that is, an operation is the execution of instructions. To say 1+1=2 is to say: “call 1+1 two”;

5) all mathematics can be reduced to summation. The hypothesis is that a) by starting with simple summation, b) and relying on the contradiction between the requirement to simplify summation and the increasing complexity of the simplification procedure itself, we can arrive at any mathematical form. This means that we can proceed from 1+1 to the most complex formulations of tensor calculus, and this will yield a solution. If we cannot arrive at it, then either it is not a mathematical object or it has no solution.

Once again. Summation enters into contradiction with the form aimed at simplifying summation, and this determines the development of summation, of mathematics as a whole. Take the summation 1 + 1 +1... When there are many units, summation no longer simplifies; complexity arises. Multiplication then appears, and 5*1 simply gives 1+1+1+1+1. Hence 5 is not something special; it is merely a representation of the operation 1+1+1+1+1, so that a more appropriate notation for 5 is 5*1. 5 is a reference to summation;

6) conversely: summation contains all mathematics within it.

2. Mathematics is the application of the idea of summation to the description of processes. It arises as follows. Elements are distinguished within the unity of the material. This division is contradictory, since there is unity, but there are also elements. This is the identification of quality. The contradiction is resolved in summation. This is the identification of quantity. Quality generates quantity—this is very important for mathematics.

Here, the qualitative and quantitative sides of the unity of the material arise. Quantity describes quality, and quality describes quantity, because quality and quantity are different moments of the same motion. The quantitative is the qualitative considered from the side of summation. The qualitative is summation considered from the side of development.

If quality is removed, there will be no quantity. “3” and “3 apples” do not mean that the former contains quantity and the latter quantity and qualities. “3” contains just as much quality. “3” is the “3” from “three apples”. When this is ignored, the result is abstract algebra.

The possibility of reducing any mathematical operation to summation is the fundamental rule of mathematics. Therefore, there are no concepts in mathematics; there is only the operation of summation and the simplification of this operation (multiplication and so on). This simplification is an abstraction over summation. Take 1+1: this is a simple notation. But as 1+1+1+… it is already a complication. A contradiction arises: summation was adopted to simplify, but it has become complicated. The contradiction is resolved in the form of multiplication.

Mathematics does not become detached from qualitative processes, even though it is the science of summation. Summation should not be an end in itself. If summation is an end in itself, mathematics becomes a one-sided exposition of its subject, which is harmful to the purposes of mathematics itself.

3. The principles of mathematics developed on the basis of summation can be transferred to logic and so on.

Subsection 1.2 Dialectical reconstruction

1. Subsection 1.1 presents a dialectical exposition of the foundations of mathematics. The traditional exposition of mathematics is non-dialectical. A transition from a non-dialectical to a dialectical exposition of mathematics is possible. This transition is called dialectical reconstruction.

Reconstruction can be applied, for example, to the following tasks: 1) reconstructing formulas; 2) solving problems; 3) extracting knowledge; 4) acquiring new knowledge; 5) code.

For dialectical reconstruction, it is important to understand that the right question is not what something means, but how it was arrived at. This applies to all knowledge.

Whereas dialectics seeks to derive the development of A into C through B, which is already present in A, through a contradiction and a leap, non-dialectical approaches simply postulate that A proceeds to C through B, that is, without checking for a contradiction or a leap. It is not enough simply to provide A, B, and C; development must be shown. Without it, we cannot move from A to B, from B to C, or, still less, from A to C. And, of course, development itself must be understood correctly.

Let us compare the approaches. The non-dialectical approach: A, B, C; we proceed from A to B, and then to C. The dialectical approach: A; B, which is identical with A and in this identity makes the transition to itself; a contradiction; the resolution of the contradiction in C. In the first case there is change, but no development; in the second there is development.

2. Dialectical reconstruction means that the object taken for reconstruction must be shown as development. A proof, equation, or problem cannot be understood from itself, as if by “staring” at it. Its development, logical or historical, must be known.

This requires the following:

1) determine what currently prevents this object from being dialectical, and what is needed to make it dialectical.

How, then, can dialectics be found? If C is given, find A, the B that develops it, the contradiction, and the leap. We trace the whole chain back to our basic understanding. If A is given, we must then seek the developing B + contradictions + C, and so on. C may not be dialectical: A or B may stand behind it, but there may be no transition, no contradiction. This must be taken into account.

This gives us an understanding of “complexity” and “unintelligibility”. The complexity of a problem is determined by the distance between what is already familiar to us and what is new. Along this entire path, we must know all the As, Bs, and contradictions. This also imposes requirements on the description of any processes and on the writing of textbooks;

2) show where the object came from and how it was arrived at. Both logical and historical approaches are possible here. This should not be confused with the situation in which the object

itself is non-dialectical and we show the development of the object itself (something like a “history of knowledge about the object”).

Thus, we seek the history of the emergence of the phenomenon that interests us. History goes “backwards” and may lead a long way back. What then? We seek the contradiction and the transition that led to what we are looking for. To discover the history of logarithms is to understand logarithms.

If the history is difficult to find, we take a logical approach: we seek a path from the abstract to the concrete, we seek development. This is not always possible. We must then bear in mind that A and B + contradictions are contained in C, so a full description of C also provides them; identifying them, however, is a matter of luck. This is precisely the kind of luck involved when we have to find the “simplest”;

3) show how the content of the process is disclosed. We must strive to disclose the process fully. Fully disclosing a process means a transition into another process contained in the original process. In disclosing a process, the following must be taken into account:

a) first, we must try to place the process itself within a dialectical account;

b) the process itself may be non-dialectical, in which case we must show how the process developed historically. In other words, if the problem itself cannot be placed within a dialectical account, the emergence of that problem can be.

One point must be considered here. We cannot convey the essence without understanding how problems are solved. We must therefore try to master the method of solution, even if it is non-dialectical. This is needed to show the history: how the problem arose and how its non-dialectical solution was arrived at.

3. Errors in exposition may include:

1) “imprecision of terminology”, when a term is not introduced and its connection with other terms is not given;

2) “lack of clarity in description”, when the description is imprecise;

3) “absence of dialectics”, when development is not shown. An error of “abstraction” is also possible here, when abstraction is carried out incorrectly. For example, the main element is chosen incorrectly, on the basis of teaching experience and the like. What, then, should be the main element? Development is the main element: we take what describes development and leave aside what does not.

Reconstructing formulas

4. A separate point about reconstructing formulas. In addition to what has been said above, two specific points must be taken into account:

1) any formula or equation can ultimately be reduced to summation, since any formula is understood only as a simplification of summation through the introduction of additional operations

that simplify summation. The difficulties that arise in mathematics therefore come solely from not knowing the operations of summation, and from nothing else.

A chain of contradictions can always be constructed from summation to any formula; likewise, any formula can be unfolded along a chain back to summation. To understand a formula is precisely to obtain such a chain;

2) a formula or equation connects the quantitative and the qualitative, and that is how it must be viewed.

Thus, we must find the development of the formula, give a qualitative as well as a quantitative description of it, and also attempt to reduce it to summation.

5. A “smart outline” must be used to reconstruct formulas. This outline is organised as follows.

On the left: A—the simplest process; B—the opposite; C—the resolution of the contradiction between A and B, which forms a new A—the simplest process; and so on.

On the right: the processes that develop the simplest process; the processes that develop the opposite, which constitutes the contradiction; the leap and the resolution of the contradiction, which itself becomes the simplest process.

In this outline, the simplest and the resolution, taken only from the left-hand side, constitute change: this is how we are accustomed to writing and thinking. The complete left- and right-hand sides taken together, however, constitute development; only this can count as understanding. To give a full description is to give a description of A in which it passes into contradiction.

This scheme provides a great deal. First, it shows how to understand C correctly, that is, to reconstruct it. Second, it shows when a failure to understand arises.

To reconstruct C means to find A and B and the transition. Contradiction is the path of the search.

An inability to understand C arises when a) it is impossible to understand C as the resolution of a contradiction; b) A and B have not been given; c) incorrect parts of A and B have been given; d) development has not been shown and, at best, only change has been shown.

Change is A and C separated from the right-hand side. Development is the complete left- and right-hand sides together. We must find hints of the right-hand side in the left; we must understand how to choose A, B, and C at all so that the “right-hand side” can be constructed.

Solving problems

6. Solving problems serves only to clarify the rules for the transition from the abstract to the concrete (from a theorem to the conditions of a particular problem), as well as the view of the world within which the problem can be posed.

7. At school or university, one often hears: “You did not solve the problem because you did not understand the topic.” But how can this be? Students often understand Ohm’s law itself perfectly well, yet cannot solve problems involving it. Students have understood Ohm’s law to the extent that it was explained to them. The secret of the problems is that nobody explained that a problem, like life, cannot be reduced to one particular law. In fact, besides Ohm’s law, there are a host of minor

laws and rules according to which reality is rearranged so that Ohm's law works within it, so that a question based on Ohm's law makes sense at all. Yet this is precisely what textbooks do not explain. It is, however, explained in solutions, in problems, but presented as the application of a law to a problem, even though it is a single whole.

From the concrete to the abstract to the concrete. This is the essence. The problem is this second concrete, whereas we arrived at the formula through the first concrete.

Students are not taught the transition from the abstract to the concrete. This transition is not explicitly contained in the abstract itself and cannot be extracted from it. Students must be taught how an abstraction is formed from the concrete and how to proceed from the abstraction to the concrete. Roughly speaking, "creating" a theory before the student's eyes is a movement from the concrete to abstraction, whereas solving a problem is a transition from abstraction to the concrete. It is precisely the failure to take this process into account that leads to a failure to understand problems. Thus, problems are the concretisation of abstraction, rather than the abstraction of the concrete. This sets the right goal: to find the rules for such a concretisation of abstraction. We do not seek the abstract in the conditions of the problem; we obtain the concrete, that is, the problem, from the abstract.

Take a typical problem: "Two travellers set out from A and B… where will they meet…?" Solving this type of problem comes down to v = s/t, that is, to the concept of speed. Yet on first encountering the problem, we must also understand that it is a problem about speed. What is central to the problem? Why speed straight away? Perhaps the weather matters. Moreover, we must also complete the statement of the problem. We must explain, for example, what "to meet" means. It means that the distance between the people is 0. But this does not happen in life: I do not stand pressed up against another person. Yet as soon as the definition is given, the problem will be solved.

Then we must do more than simply take v=s/t: we must move from this formula, from the abstract, to the concrete, that is, to the conditions of the problem. Notice that v=s/t itself says nothing about two towns or anything else in the problem.

Once again: v=s/t cannot be obtained from the conditions of the problem, but neither can v=s/t simply be taken and applied to the problem, except where the entire problem consists merely of values for v, s, t. In other words, the difficulty of "the abstract–the concrete" lies precisely in the fact that we must have further ideas for describing this concrete.

The apparent difficulty in applying Ohm's law to particular problems lies in the fact that the person has not been taught a historical approach. Historically, the concrete situation reflected in the development of knowledge about electricity in the 17th century and thereafter was reduced to an abstraction. Only by understanding this approach can we perform the reverse operation, when an abstraction must be reduced to a problem. Without this, we cannot understand how all the diversity of life is reduced to questions of electricity, and then to Ohm's law. It is like encoding and decoding. The world is encoded in Ohm's law, but we must know the algorithm of this encoding. A student cannot be considered stupid for not knowing this algorithm and not knowing how to decode the world.

Non-dialectical methods of solving problems are based on memorisation and algorithms. The school system, and education in general, are non-dialectical. Consequently, mistaken approaches to understanding the world are instilled in us from childhood. This also applies to mathematics. The inability to solve a problem is often connected precisely with such mistaken approaches. Moreover, these approaches are erroneous even within the non-dialectical approach itself.

It follows that whenever we encounter a problem we do not understand, we must ask: “Why is this not clear?” If we do not understand a problem, we must ask ourselves why. This means that the basic ideas instilled in us earlier are mistaken from the standpoint of this particular problem.

Abstractions in textbooks are often presented poorly, but solutions and practice force even idealistic descriptions to “descend” to motion (the same is true of experimental physics). History is also poorly presented in books; it too must be reconstructed dialectically. Where this is impossible historically, we must rely on dialectical logic. In any event, however, history must be checked against logic, and logic against history.

In this situation, we have to try to complete a topic as a dialectical exposition using fragments of information from different textbooks.

Section 2 History of physics

1. The history of physics can be presented only against the background of the development of industry and the productive forces, and the succession of social formations. The development of technology is an important part of this history.

The relatively simple tools of labour of the period before the Middle Ages begin to grow more complex during the Middle Ages, alongside the development of feudalism. The emergence of capitalism in the 16th century leads to the development of manufacture, of production by hand, which becomes a precondition for the creation of machinery. Machines powered by steam engines develop from the 18th century, and machines based on electric motors from the 19th century.

The connection between the development of relations of production, technology, and science can be traced through specific chains of examples. For instance, the age of geographical discoveries stimulates the development of trade and the accumulation of capital. This leads to industrial growth; the demands of navigation increase, driving the development of science. Motion becomes the central problem because people move from the idea of building enormous, static pyramids to the idea of constructing and using ships. In other words, whereas statics, the problem of equilibrium, was the central problem for ancient technology, mechanical motion becomes the central problem of the manufacturing period.

The motion of relations of production determined the motion of technology, and together they predetermined the motion of physics and mathematics as sciences. The reciprocal influence of physics and mathematics must also be taken into account.

2. In this book, we begin the history of physics with the physics of Aristotle, who held that the goal of motion was a “state of rest”. Motion in Aristotle was not dialectical.

Aristotle’s teaching persisted until the Middle Ages and was taken up and developed during them. At this time, ideas of impetus appear: of a force that the hand imparts to an object it throws. In other words, impetus was understood as the transfer of force from a moving body to a stationary one. Whereas Aristotle held that bodies tend towards rest, towards their “natural place”, the idea of the transmission of motion develops in the Middle Ages.

3. The Renaissance gives an impetus to the sciences.

4. Kepler (1571-1630) formulates his laws of motion, obtained from observations of bodies. Bodies attract one another because “kindred bodies” tend towards one another. Like Aristotle, Kepler believed that a body’s motion would cease if it were not sustained.

All the outstanding physicists of the 16th and 17th centuries address a range of questions arising from the tasks of industry. All these questions concern motion, specifically motion as change. All industrial questions were bound up with mechanics, for which a general theory had to be provided.

Galileo opposes Aristotle. In his physics, the force acting on a body is no longer tied to the body itself. But this also means that force does not depend on any particular characteristics of bodies. Equality of speeds seems to presuppose a general law for all bodies. This equality “clears” the picture.

Along the same path, Galileo chooses to describe motion itself rather than seek its causes. The idea of motion as change thus takes shape in its principal aspects. In this description, force became merely a “trigger” for motion, which was seen as change. Whereas in the Middle Ages force was conceived as pressure on a body, Galileo accepts force simply as that which changes a body’s position.

Newton does not yet adopt a coordinate system (CS), but accepts space and time as a substrate for motion. Newton’s entire theory is an attempt to fit facts into this picture of the world. For Newton, for example, “force is a directed tendency to change the state of a body”. Newton needs this proposition to set motion going. But this also means that in describing reality the emphasis shifted from motion to objects, to bodies. In this picture of the world, geometry, which describes space, now assumes the principal importance.

The idea of the ether also emerges along this path. The ether is the geometrisation, the mathematisation of description; it is the subordination of motion to geometry. Newton already connects gravitation with the density of the medium in which bodies are situated.

In the 18th century, Euler explains attraction through the elasticity of the ether; Newton had already approached this idea. The ether is simply a common coordinate system and nothing more. All its properties arise from this. In his Letters to a Princess, Euler used the ether to explain gravitation, electricity, and magnetism. This is very important. Geometrisation had reached every element. The ether became the means through which attempts were made to extend Newton’s ideas to all other phenomena. The law of conservation of energy, the change and transition of forms of motion, are an attempt in the same direction.

Newton provided all the fundamental principles of mechanics. After him, there were no substantially new ones. After Newton, these principles were merely developed. The period in which mathematics predominates over physics begins with Newton.

The contradiction that drove physics was the contradiction between motion as change and a reality that could not be reduced to change. This manifested itself through phenomena brought into consideration by the demands of industry, reflecting the development of relations of production. The study of electricity develops actively in the 18th century—so electricity must be fitted into Newton's scheme. Towards the end of the 18th century, the same trick is performed with heat.

Theories of electricity and thermal energy are attempts to resolve the contradiction between motion as change and reality, within the practical possibilities determined by the level of industrial development. This is a resolution of a contradiction in which the two opposites "coexist", but no different object arises.

As a result, time and space are coordinate systems. The essence of a CS is distance, because distance is change. Mass is a point in a coordinate system. Motion is the change of a point in a coordinate system. Forces are needed to exclude motion as the principal actor. Everything else, electricity and the rest, either conforms fully to this picture or is described through a modification of it.

In the 19th century, the contradictions of physics gathered momentum. If we look at substances and forces in the history of physics, it seems that by the 19th century everything was a complete mess. But this is wrong. Physics consistently reduced everything to motion, and motion to change. This could not proceed smoothly, and did not, but there was no mess. Take, for example, wave and corpuscular theories in optics. They were forms of change. Light was taken as change; whether through waves or molecules was a particular matter, the concrete of that abstract which defines motion. In this sense, matter was subordinated to motion, and motion to change, as follows. An object was singled out—light, say. It was taken as motion, as a process, rather than merely as a phenomenon (another example: the transition from charges to current). Motion was shown as change; what changes and how is another question, already a matter of the concrete.

In his 1832 work on magnetism, Gauss introduces the idea that units of 1) space, 2) time, and 3) mass suffice to determine all forces quantitatively. This, as it were, completes the construction of physics around a coordinate system. Forces can be observed only through their manifestation, and that manifestation is in a coordinate system. Hence the system of units of measurement. All other measurements are derived from these. These units are used to measure all forces and phenomena, not only force but also magnetism. This method prepared the unification of forces, that is, the idea of such a unification.

The idea of the transformation of forces emerges from the end of the 18th century. First, the transformation of forces into one another is demonstrated. But it must then be demonstrated that no part of the force is lost. This leads to the law of conservation. Yet the law of conservation means only that everything is motion, specifically motion in one coordinate system.

Why the law of conservation of energy, rather than of forces? Because forces were considered the fundamental characteristic of matter, and such a fundamental characteristic could not be limited in quantity. Energy was therefore adopted as the manifestation of the fundamental property of forces in concrete relations that depend on the distribution of matter in space. Thus, the focus shifts to energy. Force as a fundamental property is forgotten. Energy is taken as the motion of unified matter. This is another step towards motion as change.

We must clearly understand that this principle operates only if a coordinate system is accepted: work cannot arise within a coordinate system; it is redistributed.

Maxwell's equations show how the field governs charges. Initially, physics proceeded from material particles that generated fields. As it developed, however, greater attention was paid to fields, which now governed particles. These ideas culminated in Einstein, who wanted to replace particles with fields.

In Maxwell's electrodynamics, 1) the laws changed from CS to CS′, which cannot happen in Newton's classical physics; 2) given such changes, it was possible to determine which CS one was in.

Maxwell arrived at the conclusion that light is a wave. But a wave must propagate in some medium. The ether was chosen as that medium. Yet the ether could not be detected experimentally. There is therefore no such absolute rest in reality. At the same time, the speed of light is nevertheless constant. The approach to coordinate systems must therefore itself be changed so that this equality of the speed of light holds.

This gave rise to Einstein's theory. What led to the crisis in physics in the 20th century? Until then, the atom had been considered analogous to a material body obeying classical mechanics. This collapsed. The atom is an electromagnetic system of protons and electrons; mass increases with speed. Mass comes from the "weightless" ether. But everything consists of atoms—and that is the problem.

5. Individual electrical and magnetic phenomena had been known since antiquity. Research into them, however, developed slowly. Characteristically, magnetism was studied somewhat more, because it was important for navigation, which was important for relations of production.

Electricity, magnetism, and heat had to be brought under Newton's theory, which meant that forces had to be found. Yet these objects had no mass, so Newton could not be applied directly.

If gravitation could not be invoked, what then? These forces were then connected with "subtle matters" located in the pores of ordinary material bodies. For example, caloric was a kind of fluid within bodies. There was repulsion between its particles, and attraction between caloric and a material particle. Electrical forces were introduced in the same way. They acted at a distance, like gravity.

The idea of abandoning forces in their Newtonian sense gradually matures. It arises from Newton's own physics, in which forces merely served the coordinate system. The concept of force is not in itself necessary for motion understood as change. The only purpose of retaining this concept is to prevent the concept of "motion" from becoming independent. The distinctive feature of physics is that it describes everything as motions, as change, while leaving motion subordinate.

The principal aim of theories of electricity and heat arose from the fundamental contradiction of physics identified above. If, for you, not everything consists of motion, and you discover a kind of motion, that is, of change, you seek an "agent"—caloric, say. You can also abandon this, abandon a separate matter. You can take a single matter, but find a mechanism of change within it. In short, anything so long as matter is not motion. You have to connect matter and motion as separate things;

hence all these inventions, which, in relation to what has been said, remain variants of one thing—motion as change.

6. Mathematics serves the new ideas of physics because it is itself built on the idea of change. Mathematics contains no motion as development, but it does contain motion as change. This is how mathematics and physics connect. Differential and integral calculus, matrices, vectors, and tensors are all instruments for describing reality through physics' conception of the world as change rather than development. This is why the use of coordinate systems, which perfectly reflect this direction of thought, begins with Descartes.

7. Let us sum up. The history of the new physics contains a process of simplifying the world to motion understood as change. Physics consistently reduced the entire world to motion, and motion to change.

The world is not considered as development; it is considered only as change. Change is taken in space and time, giving a CS. This is how the geometrisation of physics occurs. The position of a body changes in this CS. Change is caused by force; force is that which causes a change in a body's position, or any other change. Precisely change, not development. In the 19th century, this will go as far as the negation of forces. But this will be done only to reduce everything to measurements, that is, to motion as change. This was creeping positivism. Interestingly, Engels overlooked it in his remark to the effect that forces were unnecessary.

The very transition to motion was a breakthrough.

Subsection 2.1 Einstein

1. This will be an important, self-contained subsection. For Einstein, with his theory, completes the development of a physics in which motion is accepted only as change. After Einstein, physics could only tread water endlessly, but that does not concern me in this book.

Einstein tried to avoid the contradiction of coordinate systems, but he ought to have acknowledged them and found a resolution of this contradiction in a higher form of motion. If a contradiction arises in motion as change, we should seek a higher form of motion as development, in which this contradiction will be resolved in unity.

2. Special relativity is built on the application of two principles—the principle of relativity and the principle of the constancy of the speed of light.

3. If CS and CS′ are given, the speed of light in these coordinates will differ when passing between them. In the 19th century, it was established that the speed of light is always the same. This means that in passing from CS to CS′, the speed of light is preserved. This requires a change, a deformation, of the coordinate systems themselves, which is what special relativity is. This sounds mystical until we consider what speed means. If we begin with the definition of speed, no mysticism remains.

4. Deriving Einstein's formula for the relation between energy and mass consists in obtaining the value of energy for CS and CS′, subtracting one value from the other, and obtaining the required formula. The law of conservation of energy must hold throughout.

5. Kinetic energy depends on the distance travelled. At the same time, it is connected with mass. The distance travelled changes between CS and CS′, which makes it possible to pull off the trick that energy equals mass.

6. Energy has inertia. This is so because of local action: energy propagates over time, from point to point. But if energy has inertia, it must also have mass. If energy had no mass, then, given its non-instantaneous transmission, a body would move, in effect, without an external influence, which is forbidden.

7. Special relativity and general relativity must be presented as pure geometry, since this exposes their essence. General relativity is the geometrisation of physics.

8. It was established that mass and weight are equivalent. Yet mass means uniform rectilinear motion, whereas weight means accelerated curvilinear motion. Consequently, if mass and weight are equal, uniform rectilinear motion must be equivalent to accelerated curvilinear motion. In other words, a CS can be chosen such that uniform rectilinear motion becomes a limiting case of non-uniform curvilinear motion, that is, inertia is replaced by gravitation.

9. Gravitation can be represented through geometry.

10. Quantum mechanics was a continuation of Einstein’s ideas.

11. Quantum mechanics uses probability, but probability is taken as non-dialectical: it contains no motion as development.

12. Whereas earlier physics simply misunderstood motion, restricting it to change, later physics, confronted with new phenomena, simply abandoned the idea of motion, substituting probability for it. Motion is now replaced by a probability amplitude.

Section 3. Dialectical physics

1. Matter must be replaced by development.

2. Matter does not have a property, quality, or form of development. Matter is development itself.

3. Development is infinity. We usually take objects and try to explain their motion (development). We must, conversely, take motion (development) and explain objects.

4. Initially, infinity is not determined by anything. This is the simplest process.

And this simplest process develops. Infinity has no boundary, but this is its boundary. Such development of infinity is the opposite of the simplest process. Taken together in their development, infinity and its opposite constitute a contradiction.

From this contradiction, infinity makes a leap into the emergence of ever more infinities. The resolution of the contradiction consists not in destroying infinity, but in giving another contradiction, another infinity.

This emergence of infinities is infinite development. We thus arrive at the conclusion that infinity is development.

Infinity is taken over the whole of infinity—this is folding. Folding means that infinity is “folded” into “this” infinity. Each folding is unique. Each folding has a probability. A leap from infinity to infinity does not require probability: the leap is determined by infinity itself. A leap from infinity to “this” infinity, however, does require probability. The laws of this probability constitute the “laws of nature”.

Folding is not scooping a handful out of the infinite; it is precisely folding infinity into “this” infinity. The probability of infinity means that if A has a probability of 5%, then for A this means existing with precisely this probability. Another property of probability is that it changes all the time, since new infinities keep arising. This can be imagined as a circle with a changing number inside it, with other circles branching out from it. Changes within do not affect the probability, since they have already been accounted for: we are taking the whole of infinity.

It is precisely the infinity of infinity that makes this entire description possible.

Qualitatively—infinity: the infinite—the boundary—the infinite.

Quantitatively—probability: the infinite is folded into “this” infinity (folding).

That is, probability is a measure for evaluating infinity, since there is no other way. Moreover, probability is not a measure of expectation, but a measure of the emergence of “this” from any.

Once again, as follows. Qualitatively: 1) infinity arises; 2) but then it has a boundary; 3) it therefore disappears; 4) but since the boundary disappears, 1) infinity appears, and so on in a circle. Quantitatively, the same process is: 1) what arises is precisely infinity, but 2) what arises is “this”, this infinity, this out of infinity. It is one and the same process, considered once in quality and once in quantity. All/this—only this is infinity: it is unbounded, yet bounded. It is therefore neither boundlessness nor boundedness; it is precisely the relation of “this to all”, of “all to this”. There are no separate “all” and “this”; they exist only in this relation.

5. Ordinary mathematics proceeds from the summation of the unit. Dialectical mathematics proceeds from folding. A folding is a unit of infinity. Such a unit has its own particular characteristics: 1) each unit is unique; 2) each indicates a probability. For example, whereas ordinary mathematics writes 1, here we write 1a, where a indicates a probability unique to each unit.

Dialectical mathematics contains only letters. The transition from letters to numerals lies in the fact that numerals already have letters even now. It is like a zero ending: it is there, but cannot be seen. That is, already 1a, rather than simply 1. Not simply 1 cm, but 1 cm in this particular situation.

“Taking over infinity” is a folding equal to a. We take again, and we have either ав (that is, including a) or в (not including it). Moreover, everything is unique. We already have two elements, even three.

6. Rules of dialectical mathematics:

1) 1+1: there is no such operation, since in my calculus there is only one, unique unit;

2) 2 can be 1+3;

3) the sign “=” is the sign for “taking over infinity”;

4) there is no sum as such; the sum shows only whether the “previous” infinity has entered here. Hence, if 1+1=2, then 1a+1в=1c or 2a, but certainly not 2. Or, more simply, 1+1 does not give two, since each unit is unique. We must remember, however, that summation is not equivalent to folding. Moreover, a folding is not a sum, but a number.

The probabilistic description can be reduced to the currently accepted description. The conditions for such a reduction must be found.

There can be no more probable and less probable in the sense that the more probable will occur. There is no probability of an event—the event is probability itself.

Subtraction from infinity yields infinity, but this is clearly a different infinity, since we are operating with development. Every operation on infinity 1) transfers infinity into another state; 2) each such state is unique, that is, has its own number.

A number is not formed by folding; the folding is itself the number. Roughly speaking, behind every 2 we must see 1+1. Moreover, 2 is not the result, but 1+1 itself. Moreover, this is not +, but precisely the whole taken together: it is precisely the moment arising from 1+… in 1+1.

A sum in dialectical mathematics is taking over infinity. The result is a folding. In ordinary mathematics, the unit indicates merely separateness from another. In my account, however, a folding is an indicator of probability. These are fundamentally different “numbers”.

A folding has uniqueness; hence a, в, c, and so on reflect precisely uniqueness. Uniqueness itself is the principal quantitative reflection of infinity. Yet the unique is still taken over something, over infinity.

In my account, to have a probability of 56 percent directly means to exist as a “probability of 56 percent”.

7. The mathematics of infinity does not measure the length of a line segment; it calculates the probability of that segment.

A table does not consist of infinity; the table is infinity itself. This applies to all its properties, qualities, and so on. A table differs from a chair in the “probability” of this particular infinity. The table has a probability of 5, and the chair of 10.

8. The mathematics of infinity must be used to show, for example, what electricity is. Specifically, quantitatively—which means that an “adapter” from accepted data to infinity is needed.

9. Motion is development. In dialectical physics, motion is understood precisely as development. What is the difference between development and change? Suppose there is a body. Let it move from point A to B. Change tells us: the body moved from A to B. But now suppose that the body at A and the body at B are not the same body, even if they are completely identical. What then? Change then gives us nothing with which to describe the situation. That is the point. Development is needed here; change alone is insufficient. Past physics assumed the identity of two bodies at different points in space and time. Dialectical physics denies this identity, and also denies space and time. Once development is given—precisely given—space and time are not required.

When it is assumed that motion reduces to change, it is assumed that position can be determined only through a frame of reference. But this is not so. Position can be determined through history,

through development, through motion. Since change also occurs in motion, it is of course possible to isolate change alone and enclose it in a CS. But this does not alter the fact that such change is merely an abstraction from actual motion.

The difference between development and change can be understood as follows. When it is said that on a ship from which we cannot see any objects outside, we cannot determine whether we are moving, what is meant is change, not development. We observe no change, and it is therefore concluded that there is no motion either. But this is not so. The ship moves because it has its history, from the first tree sawn to unloading in port. We must not give in to the desire to see time in this. What must be clearly seen here is precisely development.

We must not take an object and explain its motion; we must take motion and explain the object.

10. The discreteness of existence. Not the discreteness of bodies, space, or energy, but of existence itself. It is assumed that a body exists, with time and space around it. But the question must be posed about existence itself. At every “instant”, the question of existence is posed and resolved: existence is discrete, not monolithic. Something exists in space and time. This is where people begin, whereas they should begin with existence itself. A ray of light at A and a ray of light at B are therefore different rays, connected by development, but not something merely situated in space and time as in a box.

Section 4. Education

1. Education derives from capitalism. Education today therefore 1) does not provide a dialectical understanding of the world; 2) is based on caste in its very essence, which manifests itself in books, in methods of exposition, and in science itself. Education that does not aim to be education for everyone does not attempt to present knowledge simply and clearly.

2. Education must be dialectical. And education must be directed towards mastering dialectics, that is, comprehending the world through development.

The dialectical approach to education proceeds from the following: 1) our brain is dialectical from the outset; it perceives the world dialectically, even when this occurs unconsciously; 2) a person’s education, a person’s acquisition of knowledge, is a process of dialectical development; 3) non-dialectical exposition and teaching, running counter to the dialectical nature of the human being, the brain, and the learning process, only complicate learning; 4) learning must begin with dialectics.

The non-dialectical approach makes knowledge appear as arbitrarily cut pieces that may—but need not even do this—show change, but not development. How engaging courses and books are depends on the skill with which these pieces are cut, but this does nothing to help understanding.

The dialectical approach should simplify memorisation. Why do we remember poorly? Because, at best, we memorise change, rather than development.

3. Education should strive to make acquiring knowledge simple. Simplicity is needed for speed and for the volume of knowledge; it is not an end in itself.

4. Studying a subject involves a transition from the abstract to the concrete. We acquire abstract knowledge that must be made concrete. If difficulties arise, we should turn to 1) the history of the subject; 2) classic works on the subject, which themselves reflect its history.

5. Do not try to memorise something; strive to understand, to make the motion clear to yourself.

6. Do not be afraid of forgetting something. Forgetting as a principle is an important element of learning. What we forget is not what matters most; what matters, we do not forget. What matters is another question. Forgetting is not the same as selecting knowledge.

7. Learning should proceed in “layers”, in passes through knowledge. Knowledge is acquired in strata, that is, by going through one topic several times and in different ways. This fully accords with dialectics: at each level, a particular stratum of knowledge is acquired, and there is a transition from abstraction to the concrete.

Complete mastery on each pass is not required. Connections are established gradually. The level of complexity must be increased on each pass. These passes work like a magnet. We go through the material, acquire something, and establish connections. We do not necessarily have to acquire something central. For now, what matters is acquiring knowledge. With each pass, the magnet grows stronger. The whole body of knowledge is in our head; it is simply not accessible to us in its entirety, whereas the magnet, the strengthening of connections, understanding, and the very mechanism of understanding grow. In its essence, this is partly a process of forming abstract-concrete concepts.

Such learning is also needed because we learn several things at once. Otherwise it is boring, although boredom, like fatigue, is not an enemy and can quite well serve as a guide. Learning itself must change direction while holding to one goal. There must be variety.

8. Choosing books is an important element of education. To choose books, it is important to understand that books are written under the dictates of capitalism.

Textbooks and books are not written on the assumption that everyone should grasp the knowledge. As a rule, the assumption is that textbooks are needed by students to whom transitions in formulas and additional concepts can be explained in lectures. As a rule, these additional concepts are the crux of the matter; without them, knowledge appears abstract, without the concrete. These additional concepts need to be discussed in greater detail.

Books can be read, or they can be skimmed.

Skimming a book differs from reading it in that skimming only extracts data from the book for the picture of the world, the “scheme”, the processes that have already been grasped. Reading also includes making sense of things, reflection, and constructing that very picture of the world.

Skimming will therefore be effective to the extent that you already have a grasp of the processes you are reading about. This does not exclude the possibility of gaining knowledge while skimming.

We should take from a book only what we came to it for. The following can be omitted when skimming a book:

1) mentions of other books and authors, accounts of their ideas, since these are already known. This does not, of course, apply when the specific aim is to learn what another author thinks;

2) the author's disputes with other authors or ideas. Again, this does not apply when it is precisely the disputes that interest us;

3) the author's opinions and evaluations. Such evaluations are most likely not dialectical;

4) everything that does not concern change or development.

The fundamental problem is that books contain no dialectics. Yet what we need to find is precisely dialectics. This is addressed through dialectical reconstruction.

This gives us an understanding of the fundamental limitation of ordinary books: they present change, but not development. And it gives us a task: to seek change, but construct development from it. We can also use already known laws and principles of development that we have learnt from other books. For example, we know that creative work is determined by class and circumstances, so when skimming, we attend precisely to these facts.

For example, in books we can seek transitions that clearly indicate changes. We need transitions, not simply chronology. For example: finished school and chose a particular university, rather than simply finished school at a particular time and entered university at a particular time. This is not yet development, but it is already change.

Enumeration differs from change in that it does not matter in enumeration whether one thing differs from another: a simple "a, a, a, a…" is also an enumeration. Change, however, is "а, б, в…". This is what initially interests us, since we cannot find development in these books anyway.

9. A scientific text should be constructed as follows: motion (how we arrived here), examples (even if they repeat the motion), and where we will proceed from here. Added to this are the tricks that, consciously or otherwise, make understanding difficult. Each text should be no longer than a specified limit. This should not, however, interfere with completeness of exposition. A text should describe motion and show its interconnection with others, and do so non-mechanically: simple tags are insufficient.

10. Formulas should be written as a+в=c. In lengthy arguments, anchors from earlier arguments should be repeated. The frequency should be appropriate to the context.

11. Difficult words should additionally be given an easy explanation in two or three words. Such an easy explanation provides an anchor for the brain, for recall.

12. Knowledge is currently acquired in pieces, through definitions. It is not presented historically. It is presented through change, that is, through a transition from one thing to another, rather than through development.

Section 5. Artificial intelligence

1. AI cannot be created as a complete analogue of a human being.

First of all, AI based on computers cannot escape the limits of binary "thinking". The world surrounding AI, like the world surrounding us, cannot be reduced to 0 and 1. The world is dialectical; human reason is dialectical. The computer is non-dialectical; AI is non-dialectical. This

fundamental difference cannot be overcome. AI is incapable of dialectics, incapable of providing concepts or showing development.

For the same reason, AI will never be able to gain access to the whole body of knowledge. Universality is an important component of thinking. A compass can only draw a circle, but the human hand can draw both a circle and a square. The compass will make the circle perfect, whereas the human will not, yet the human has the advantage precisely through universality. It is important to understand that even imperfection is an important component of knowing the world, just as forgetting is an important part of understanding.

Why does ordinary machine learning work, even though it is mistaken? Because abstractions are, by their nature, rooted in reality. By guessing abstractions, we can therefore also guess reality, the concrete.

2. When I say, “The apple is tasty”, and a computer writes, “The apple is tasty”, these are different things. What is the difference? My judgement, my concept, is abstract-concrete: it contains development, motion, whereas the computer does not make a judgement at all; this is not a concept at all. There is change here, but no development. The deception is the same as in physics, where change is substituted for development. If this is difficult to understand, we need only imagine the actual history of what preceded “the apple is tasty” for the human and for the computer. AI can grasp change, but it cannot grasp development.

3. Only a science and a historical moment that disregard dialectics could have posed the question of the possibility of AI at the level of human reason.

4. Contemporary machine learning, misunderstanding AI and its prospects, develops and uses methods that prevent us from “squeezing” everything possible out of AI.

Contemporary machine-learning methods thus generalise incorrectly. For example, the decision-tree method divides objects according to a feature and attempts to find a common feature. But this is the wrong path.

Take books. Dividing them by page count does not create a criterion for evaluating a book. We must establish the history of book evaluations and identify what drove that history: books became thicker and thicker, evaluations declined, and so on.

The following must be understood in the same sense. Defining a circle through the procedure for constructing it provides a correct concept. Defining it through properties (equidistance from a point) provides an imprecise concept, without development.

In the same sense, neural networks attempt to simulate information being taken from an indeterminate set and to find motion. Neural networks are an unconscious solution to the task of dialectics, a solution that fails to achieve its goal. The principal weakness of neural networks is that it is not the brain that thinks, but the person, albeit through the brain. This is the essence: such thinking extends beyond mere processes in the brain. A person sees not with their eyes, but with the whole of human experience, the whole of human history—not only that of a particular person, but the history of society.

5. Artificial intelligence cannot understand reality dialectically. AI is incapable of finding development or showing motion. AI is, however, capable of at least two things: 1) AI can "measure" change in data; 2) AI can process data quickly. Change is not motion, since it is not development. Yet change is an element of development; development can be considered from the side of change.

Taking these two characteristics of AI, we can formulate its task: to provide materials for a dialectical solution. AI cannot rise above this task, but the task is itself enormous and important.

6. To achieve the stated aim, dialectical learning must be established. Dialectical learning means an interaction between a human and AI whose result, on the AI side, is knowledge, and on the human side, a dialectical understanding of such features.

The aim of dialectical learning is to teach dialectical analysis.

To achieve this aim, it is important to resolve the problem of the relation between change and development.

7. AI fundamentally works with probability. This probability, however, must be reworked.

Why can ordinary probability not be used? Because it has a key deficiency: probability is used to evaluate the result from the starting point, without taking into account the execution itself, the transition from the starting point to the result. We take the ratio a/b or c/b before the "choice" that will yield a or c is made. If the world changes between the balls being in the bag and a ball being selected, probability does not account for this. Moreover, for probability, a and c are separated from one another, whereas dialectics has no such rigid separation.

The problem with probability is that it works once opposites have already been found—say, "a bag of balls" and "a blue ball has been selected". AI, however, should help find the opposites themselves, and this will yield a different, simpler and more accurate evaluation of probability.

It is important to bear in mind that a blue ball in the hand is not the realisation of 4/6, where exactly 4 balls are blue. It is simply a blue ball. The meaning of 4/6 is that if we draw balls endlessly, we will obtain blue balls more often.

8. In constructing "concrete-universal concepts", we realise the crucial aim of proceeding "from the whole to the conditions of its possibility". Our aim is not to find how the target variable depends on a set of variables in a particular row. Our aim is to find how the whole, that is, the entire dataset, is formed from the conditions of possibility of that whole, to find the conditions of that possibility.

9. The same dialectics that we used for mathematics is used for AI. Here, however, it must be used specifically for the purposes set for AI. To do this, we must understand that AI is limited to describing the world through mathematics that is not dialectical. This means that we must understand how mathematics approximates dialectical reality, that is, how change approximates development.

The AI algorithm

10. Current AI is used to collect data, process it, and provide software support. New AI is trained on the data collected by current AI.

Information is currently contained in textual and graphical documents, with different formats. A common method for collecting, describing, and transforming this information is needed.

11. In implementing the AI algorithm, we proceed from the conception of motion as development formulated in the section on Dialectics, and from the visualisation of that conception through the “smart outline”.

With this in mind, the aim of AI training is to identify opposites that can be considered as contradictions. This is how AI participates in human reflection. AI provides the human with opposites; the human checks them for contradiction and constructs development. In the terms of the “smart outline”, AI provides opposites, while the human determines whether they correspond to A and B, and also establishes the leap to C. The designations A and B are retained within the algorithm, but it must be understood that their final status as contradictions is determined by the human.

AI can distinguish 0 from 1, that is, it can establish an opposite. AI can therefore distinguish A, B, and C, but cannot establish contradictions or a leap, that is, establish development.

12. The algorithm is implemented through the following steps:

“1. The world is taken as a single infinite process undergoing development and containing other processes nested within one another.

1.1 When one process passes into another process, this is what is called a process. A process can be specified only as the transition of one process into another process.

1.2 All individual processes exist within this single process. Any process is connected with other processes in the infinite development of the world.

2. Processes change.

2.1 ‘Processes change’ means that the simplest process changes in a way accepted in practice or that may follow from practice.

2.2 A process is considered the simplest when it simultaneously 1) is itself the resolution of a previously arisen contradiction; 2) can develop.

3. Processes develop in iterations.

3.1 An iteration is considered a finite state of an infinite process. At each iteration, we obtain a finite set of developing processes, but consider it an expression of the infinite world undergoing development. Each iteration is therefore simultaneously complete and incomplete. Complete, because a particular set of developing processes has already been obtained. Incomplete, because its development is in principle not exhausted.

3.2 The simplest process changes in an infinite world. The full development of the simplest process is therefore infinity. For each and every simplest process, its full development is infinity. In this sense, simplest processes are identical in infinity, and this constitutes the unity of the world.

3.3 In its infinity, the world is one; hence, over infinity, any process is connected with any process.

3.4 Infinity taken in a particular iteration is the simplest process. The converse is also true: the simplest process is infinity taken in a particular iteration.

3.5 At every iteration after the first, the number of developing processes changes: there may be more or fewer than at the previous iteration. The number of developing processes at each iteration must be unique across all iterations.

3.6 At every iteration after the first, the composition of developing processes must differ from the composition at all previous iterations, that is, it differs from all developing processes.

3.7 At the first iteration, the developing process follows from the development—and specifically the development—of the simplest process and only the simplest process. At the second and subsequent iterations, the developing process follows from the development—and specifically the development—of the simplest process together with all previous iterations.

3.8 At each iteration: 1) developing processes are mathematically compared with the simplest process; 2) developing processes are mathematically compared with one another. The comparison concerns both the content of the developing processes and the path traversed to reach the developing process.

3.9 Each iteration is considered unique, which must be ensured mathematically. Uniqueness means that each iteration adds information about infinity. In an equation, this is reflected as the property of uniqueness of each equation.

3.10 Iterations constitute motion, rather than a sequence of motionless states. This means that the current iteration already is the next within motion itself. This can be termed the ‘generation of developing processes’. It occurs as follows: the simplest process already is what it is, that is, ‘the one’, but also contains what it is not, that is, ‘the other’. The simplest process presents itself as the identity of ‘the one’ and ‘the other’, through which the transition into ‘the other’ takes place. This ‘other’ is called a developing process. Each developing process generates subsequent developing processes as the simplest process generated a developing process.

4. The tendency of the generation (iterations) of developing processes towards infinity constitutes the opposite of the simplest process.

4.1 The opposite understood in this way can, in its reverse unfolding, replace the simplest process.

5. The simplest process and the opposite process, taken in the unity of their development, constitute a contradiction.

5.1 The resolution of the contradiction (sublation) takes place through a leap. This means that the contradiction is resolved without the identity of ‘the one’ and ‘the other’. In this case, only the unity of the world and the determinacy of development connect the simplest process and the resolution of the contradiction.

5.2 The resolution of the contradiction means that either a state has been found in which the contradiction disappears, or a state has been found in which the contradiction is replaced by another.”

13. Mathematically, the search for opposites can be described as follows:

1+1+1+1… is disclosure

1+1=2… is merely a counter of properties

1+1=3… is a search for an opposite, and moreover

1+1=3, but also 1+1+1+1=3, and 1+1+1+1+1+1+1=3… summation does not change the value on the right,

yet 1+1=2, 1+1+1+1=4, 1+1+1+1+1+1=6.

Probability can then be specified as follows:

1+1=2, 1+1=3, which gives 2/3… this is probability.

To apply this probability, rules must be established for specifying 3: that is, why 1+1=3 rather than 1+1=4. We must also determine what we count as 1, what “fact” we are speaking of. This must be a dialectical fact: not balls in a bag, but the situation “a ball in a bag”.

Probability in this case is understood as the probability of an opposite arising as A is disclosed with event B.

14. Data today are pieces between which we seek connections. Yet the connections themselves should become the “pieces”. This will make it possible to compress data and simplify its processing.

Any data that are studied or used to train AI must be restructured so that they correspond as closely as possible to the idea of development.

The correct algorithm for preparing data is: 1) make the data move; 2) find development in that motion (the simplest, the abstract-concrete); 3) discard what is unnecessary.

15. The following additional ideas may be useful in developing AI:

1) outliers. If problems are solved endlessly, the correct solution can be selected by treating it as an outlier. Moreover, for a machine, outliers are what do not fit the algorithm, while the machine’s algorithm cannot fully realise dialectics. Outliers may therefore potentially indicate precisely a dialectical circumstance;

2) reverse programming. The meaning of reverse programming is as follows. AI, with its 0 and 1, cannot find contradictions or a leap. It can, however, find the very fact of a contradiction—this is where AI breaks down. Reverse programming precisely makes AI ‘break down’;

3) data compression. If we take the ‘smart outline’, data can be compressed in two ways: a) we can take only the left-hand side of the outline. In this case, the right-hand side will be the expansion of the data; b) we can take only C. Not just any three ABC can be compressed, since not just any three yield dialectics.